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Task-4 : (Short Question Answer)

1) First, I would convert the rover's GPS Latitude and longitude into local East-North coordinates relative to the given grid origin. Next, I would determine the grid's rotation angle using its edges or known reference points. Since the grid is not aligned with the compass directions, I would apply a 2D rotation transformation to convert the East-North coordinates into the grid's coordinate system. Finally, I would use the grid boundaries to identify the rover's cell.

Justification: GPS provides the rover's geographical position, but latitude and longitude alone do not directly tell us which cell it occupies in a rotational grid. Therefore, converting the coordinates and accounting for the grid's rotation is necessary for accurate localization. The rover's facing direction is irrelevant because

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its position can be determind independently of its orientation.

2) (Linux USB serial port conflict)

⇒ The problem may be caused by both the GPS and IMU nodes using the same USB serial device path, such as /dev/ttyUSB0. Since a serial port is normally accessed by only one process at a time, launching the IMU node may cause a conflict and stop the GPS node from receiving data.

The issue may switch after reboot because Linux can assign USB device names in a different order.

To fix this, I would assign each sensor a unique and stable device path using its

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USB serial number or Linux udev rules. For example, I would use seperate paths under /dev/serial/by-id/ for the GPS and IMU and update their respective configurations.

Justification: Using temporary device names such as /dev/ttyUSB0 is unreliable because their assignments can change after reboot, stable device paths identify each physical sensor consistently. this prevents both nodes from accidentally accessing the same port and ensures that each sensor communicates with the correct program.

